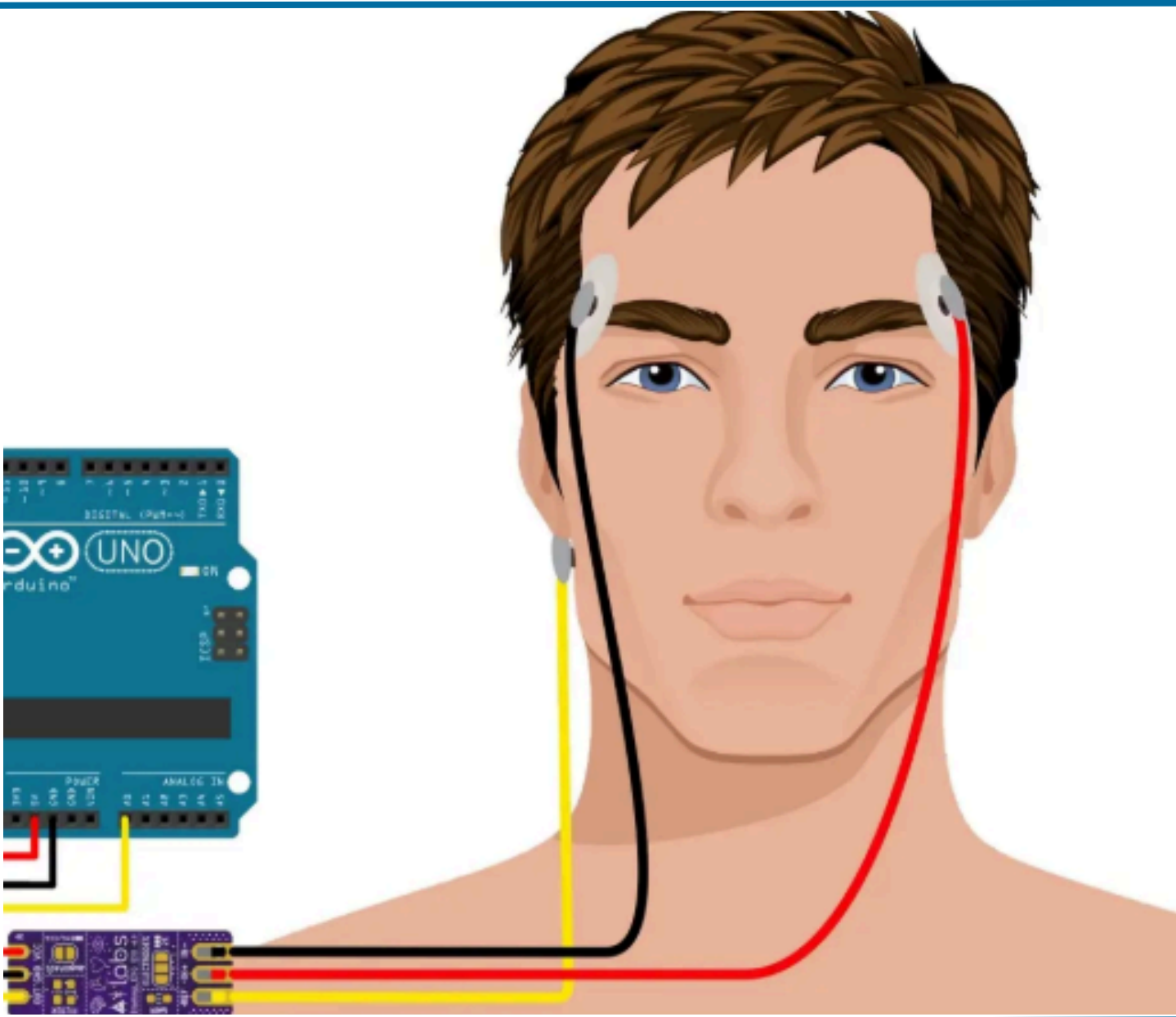
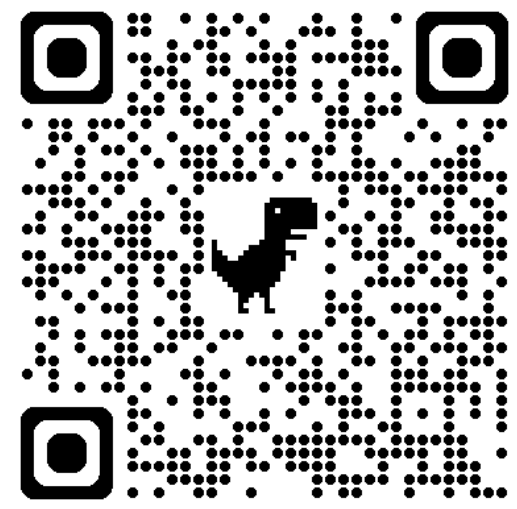




Introduction and Abstract

Brain-computer interfaces help people send messages using brain signals. Our project explores how these EEG signals can be recorded and turned into actions on a computer.

Methodology

Signal Acquisition: It starts with collecting brain signals using simple electrodes on different body parts.

Analog Signal Conditioning:

These signals are connected to band-pass filters to remove any noise. Since the voltage generated from these signals is low, an amplifier is also added

Digital Signal Processing: A

microcontroller (ESP32) is used to convert analog signal to digital data. It performs moving-average filtering and peak detection for blinking
In the case of muscle movement, envelope detection algorithm is implemented

Results/Findings

Blink detection: An action on the computer gets triggered for every eye-blink

Muscle Movement: Based on the wrist movement on either arm, the computer cursor moves in the respective direction

Conclusions

This project demonstrates the working of a low-cost brain-computer interface (BCI) system using custom-built signal conditioning circuits and an ESP32 microcontroller